DE HAVILLAND AIRCRAFT OF CANADA LIMITED

TEMPORARY REVISION

PSM: 1-64-2 **AIRCRAFT: DHC-6** **SERIES: 400**

PUBLICATION: MAINTENANCE MANUAL **TEMP. REVISION NO.: TR77-1**

Insert in 77-10-00 before Page 1

TORQUE PRESSURE INDICATING SYSTEM - GENERAL DATA

REASON FOR ISSUE:

To introduce Engine Torque Transducer Removal, Installation, Calibration and Testing procedures.

ACTION:

In Chapter 77, Section 77-10-00, ignore existing Page 1 and insert revised pages as shown on Pages 2 to 4 of this Temporary Revision.

-Continued overleaf-

RECORDING INSTRUCTIONS:

Record this TEMPORARY REVISION NO. and the relevant information in the appropriate columns of the TEMPORARY REVISION INDEX provided at the front of the manual.

Torque Pressure Indicating System – General Data

1. General

The torque pressure indicating system consists essentially of a pressure transmitter mounted on the engine reduction gear case. Torque pressure indication is displayed in the engine parameters display, provided in the upper inboard 1/6th of the Primary Flight Display. The power supply for the indicating system is 28 Volt DC with 5–ampere circuit breakers, marked TORQUE PRESSURE L and TORQUE PRESSURE R.

A. Torque Pressure Indication

Torque pressure (TRQ PSI) is indicated with a pointer type display with integer values beneath each display. Units are displayed in psi.

B. Pressure Transmitter

The torque pressure transmitter is mounted on the engine reduction gear case and is used to transmit a pressure differential, obtained from two tappings on the reduction gear case, which is supplied to the DAU for conversion to digital data for the Apex system.

2. Removal/Installation

WARNING FAILURE TO OBEY ALL THE SAFETY PRECAUTIONS WHEN DOING MAINTENANCE ON OR NEAR ELECTRICAL/ELECTRONIC EQUIPMENT MAY RESULT IN INJURY TO PERSONNEL AND/OR DAMAGE TO THE AIRCRAFT. REFER TO 24-00-00, ELECTRICAL POWER — GENERAL DATA, PARA 2.A.

A. Removal of the Torque Transducer

1) Disconnect Aircraft Power (Refer to Maintenance Manual, Chapter 24).

2) Remove applicable electrical connector, P175 (left engine), or P176 (right engine).

3) Disconnect hose assy from torque transducer.

4) Remove AN815-4D union from torque transducer and retain for re-installation. Discard O-ring.

5) Remove torque transducer from engine gearbox adapter.

B. Installation of the Torque Transducer

Note Testing of the transducer must be accomplished prior to installation of the transducer into the engine.

1) Test transducer (Refer to Para 3.A).

2) Install AS3209-213 O'ring onto the transducer. Lubricate O-ring with engine oil prior to installation.

Note Alternate O-ring part number NAS617-12.

3) Install the transducer into the gearbox adapter. Torque the transducer to 68 - 75 in/lbs, and safety.

4) Install hose assy (PN 601002-4-0240) to the union of the transducer. Torque as per Maintenance Manual, Chapter 20-20-01.

5) Connect the electrical connector P175 (left engine), and/or P176 (right engine).

6) Do Engine Leak Test Run (Refer to Maintenance Manual, Chapter 71-00-00, Page 14). Verify no oil leaks are present after installation.

3. Adjustment/Test

Note There are no adjustments that can be made on Torque Transducer part number APTE-548-2250-70D.

A. Component Test

1) Install AS3209-04 O-ring on AN815-4D union. Lubricate O-ring with engine oil prior to installation.

Note Alternate O-ring part number NAS617-4.

2) Install union to transducer. Torque union as per Maintenance Manual, Chapter 20-20-01.

3) Install the hose assy to the union of the transducer. Torque as per Maintenance Manual, Chapter 20-20-01.

4) Connect the electrical connector P175 (left engine), and/or P176 (right engine).

5) Attach Model 2311F Pressure Tester, or equivalent, to the gearbox input side of the transducer.

6) Apply 28 VDC power to aircraft and monitor engine torque parameters on APEX MFD screen.

7) Record indications at each reference pressure on Figure 1 as follows:

a) No input pressure (0%). Record the value.

b) Increase the pressure to 50% of FS and record the value.

c) Increase the pressure to 100% of FS and record the value.

Note The parameter display limits on Apex may not allow for display of the 100% values. Modify the 100% value in this case to the highest reading displayed on Apex without having the value pinned at its upper most value.

8) Confirm the test points are within the Pass criteria band.

9) Remove pressure tester from transducer.

10) Remove electrical connector (P175 and/or P176) and disconnect hose assy from transducer prior to installation of transducer into the engine gearbox.

Test Pressure Output (Note: Values indicate % of full scale.)		
0%	**50%**	**100%**
Nom: 0 PSI Actual: _____ Pass Criteria: -0.7 to 0.7 PSI	Nom: 35 PSI Actual: _____ Pass Criteria: 34.3 to 35.7 PSI	Nom: 70 PSI Actual: _____ Pass Criteria: 69.3 to 70.7 PSI

Tolerances – Torque Pressure Indication
Figure 1

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